Introduction à la Programmation et l'Algorithmique

CHRISTELLE CAILLOUET – CHRISTELLE.CAILLOUET@UNIV-COTEDAZUR.FR

DINO LOPEZ PACHECO – DINO.LOPEZ@UNIV-COTEDAZUR.FR

GDB – The GNU Debugger

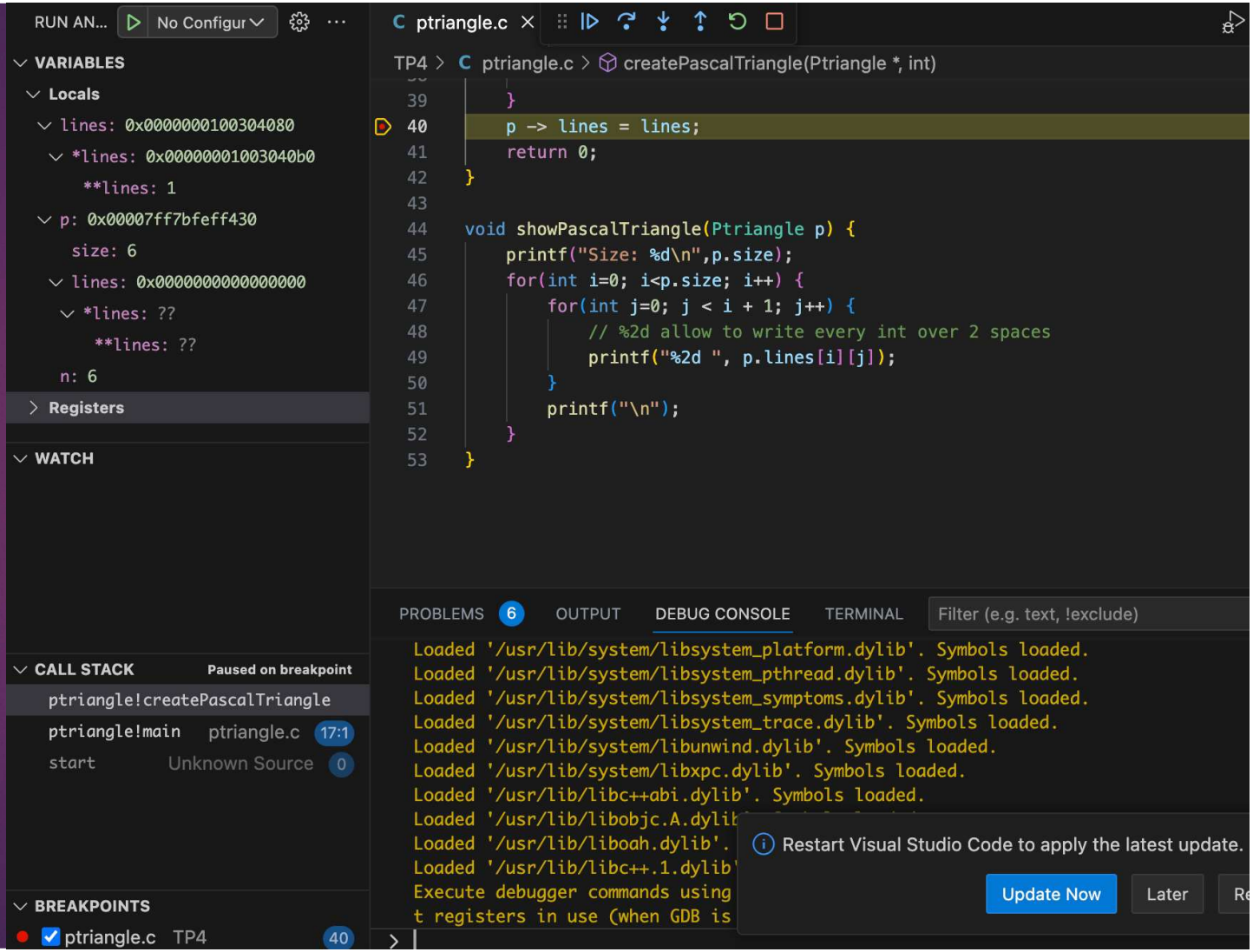
Debuggers and GDB

- Programs might crash or give unexpected results
  - At any point of its lifetime!
- Analyze and inspect the program at a particular time of its execution can help a lot
- Debuggers are the tools that you need to inspect your programs
- GDB, the GNU Project Debugger, is one of the preferred debugger to analyze C programs
  - Support many other programming languages (Go, Rust, …)

- GDB can be used
  - Natively, in text mode – a lot of docs in the net
    - https://web.eecs.umich.edu/~sugih/pointers/gdbQS.html#:~:text=Setting%20breakpoints%20A%20breakpoint%20is,at%20line%2055%20of%20main.
    - http://perso.ens-lyon.fr/daniel.hirschkoff/C_Caml/docs/doc_gdb.pdf
    - https://www.cs.umd.edu/~srhuang/teaching/cmsc212/gdb-tutorial-handout.pdf
  - Or graphically – plugins on your favorite editor

GDB in Visual Studio Code

https://code.visualstudio.com/docs/cpp/config-linux



Compile with GDB support

- Machine code aren't easily debugged by default
  - No easy way to stop the execution of a normal program
- To debug a program, one needs to insert special "capabilities" during compile time
  - Usually, with GCC you do "gcc –o welcome welvome.c"
  - Compilation with GDB support "gcc -o welcome -g welcome.c"

Launching and breaking a program

- Launch your program debugging
  - $ gdb welcome
  - Then, you should see the GDB prompt
    - (gdb)
- Run a program
  - (gdb) run
  - This will run the program until a breakpoint is found
  - By default, there is no breakpoint
- Break the program execution
  - (gdb) break 16
  - Or break in a function (“int myfunc(int var);”) : (gdb) break myfunc

Program inspection

- Once your program stops, you can inspect your variables
  - (gdb) print myvar
- And execute the program again, until a breakpoint is found if any
  - (gdb) continue
- Or, you can execute line by line your program
  - Execute the next line (but if the next line is a function call, you don't enter the function)
    - (gdb) next
  - Execute the next line. If the next line is a function you coded, you enter the function to eventually execute line by line
    - (gdb) step

Some useful GDB commands

- Print in hexadecimal format
  - print/x myvar
- Print a struct pointer variable
  - print mystruct->myvar1
- List your breakpoints
  - info breakpoint
- Delete a breakpoint
  - Assuming you want to remove breakpoint # 1 (as listed by "info breakpoint")
  - delete 1